Chapter II

Linear Time Invariant Systems

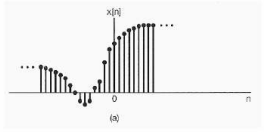
- Discrete-Time Linear Time-Invariant Systems
  - The convolution sum
- Continuous-Time Linear Time-Invariant Systems
  - The convolution integral
- Properties of Linear Time-Invariant Systems
- Causal Linear Time-Invariant Systems Described by Differential & Difference Equations
- Singularity Functions

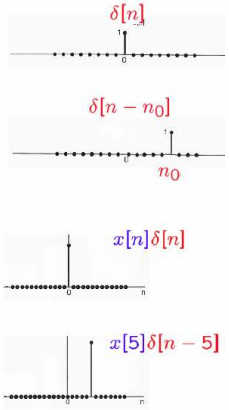
- Sample by Unit Impulse
  - For x[n]

$$x[n]\delta[n] = x[0]\delta[n]$$

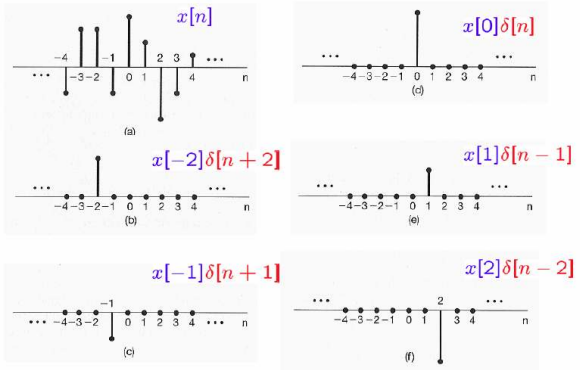
  - More generally,

$$x[n]\delta[n-n_0] = x[n_0]\delta[n-n_0]$$





- Representation of DT Signals by Impulses:



- **Representation of DT Signals by Impulses:**
  - More generally,

$$x[n] = \cdots + x[-3]\delta[n+3] + x[-2]\delta[n+2]$$
$$+ x[-1]\delta[n+1] + x[0]\delta[n] + x[1]\delta[n-1]$$
$$+ x[2]\delta[n-2] + x[3]\delta[n-3] + \cdots$$
$$= \sum_{k=-\infty}^{+\infty} x[k]\delta[n-k]$$

  - The sifting property of the DT unit impulse
  - x[n] = a superposition of scaled versions of shifted unit impulses δ[n-k]

- **DT Unit Impulse Response & Convolution Sum:**

$$\text{input} \longrightarrow \boxed{\text{Linear System}} \longrightarrow \text{output}$$

$$\delta[n] \longrightarrow \boxed{\text{Linear System}} \longrightarrow h_0[n]$$
$$\delta[n-1] \longrightarrow \boxed{\text{Linear System}} \longrightarrow h_1[n]$$
$$\delta[n-2] \longrightarrow \boxed{\text{Linear System}} \longrightarrow h_2[n]$$
$$\vdots$$
$$\delta[n-k] \longrightarrow \boxed{\text{Linear System}} \longrightarrow h_k[n]$$

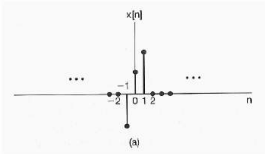
- **DT Unit Impulse Response & Convolution Sum:**

$$x[n] \longrightarrow \boxed{\text{Linear System}} \longrightarrow y[n]$$

$$x[0]\cdot\delta[n] \longrightarrow \boxed{\text{Linear System}} \longrightarrow h_0[n]\cdot x[0]$$
$$x[1]\cdot\delta[n-1] \longrightarrow \boxed{\text{Linear System}} \longrightarrow h_1[n]\cdot x[1]$$
$$x[2]\cdot\delta[n-2] \longrightarrow \boxed{\text{Linear System}} \longrightarrow h_2[n]\cdot x[2]$$
$$\vdots$$
$$x[k]\cdot\delta[n-k] \longrightarrow \boxed{\text{Linear System}} \longrightarrow h_k[n]\cdot x[k]$$

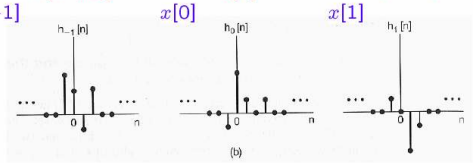
$$x[n] = \sum_{k=-\infty}^{+\infty} x[k]\delta[n-k] \quad \Longrightarrow \quad y[n] = \sum_{k=-\infty}^{+\infty} x[k]h_k[n]$$

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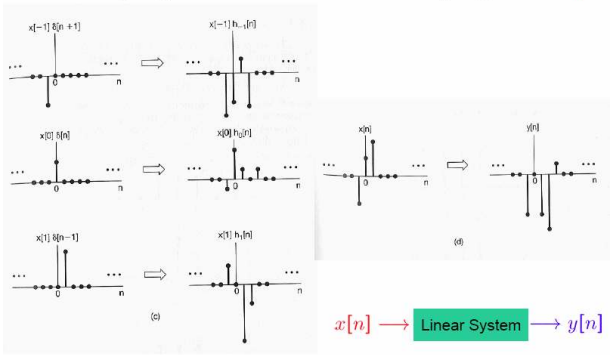


$\delta[n+1]$ $\quad$ $\delta[n]$ $\quad$ $\delta[n-1]$

$x[-1]$ $\quad$ $x[0]$ $\quad$ $x[1]$



$$x[n] = \sum_{k=-\infty}^{+\infty} x[k]\delta[n-k] \quad \Longrightarrow \quad y[n] = \sum_{k=-\infty}^{+\infty} x[k]h_k[n]$$



$x[n] \longrightarrow$ Linear System $\longrightarrow y[n]$

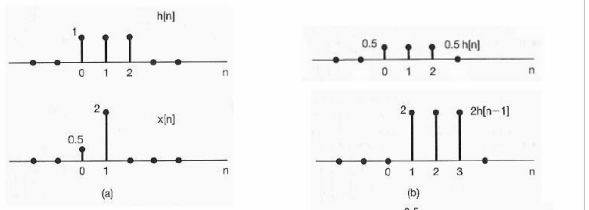
- If the linear system is also time-invariant
  - Then, $h_k[n] = h_0[n-k] = h[n-k]$
- Hence, for an LTI system,

$$y[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k] = \sum_{k=-\infty}^{+\infty} x[n-k]h[k]$$

  - Known as the convolution of x[n] & h[n]
  - Referred to as the convolution sum or superposition sum
- Symbolically, $y[n] = x[n] * h[n] = h[n] * x[n]$

- Example 2.1: $y[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k]$ $\quad x[n] \longrightarrow h[n] \longrightarrow y[n]$

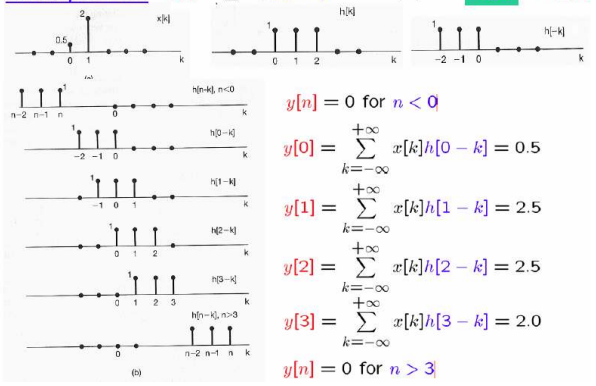
$$= \cdots + x[-1]h[n+1] + x[0]h[n] + x[1]h[n-1] + x[2]h[n-2] + \cdots$$



$$y[n] = x[0]h[n-0] + x[1]h[n-1]$$

$$= 0.5h[n] + 2h[n-1]$$

- Example 2.2: $y[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k]$ $\quad x[n] \longrightarrow h[n] \longrightarrow y[n]$



$y[n] = 0$ for $n < 0$

$$y[0] = \sum_{k=-\infty}^{+\infty} x[k]h[0-k] = 0.5$$

$$y[1] = \sum_{k=-\infty}^{+\infty} x[k]h[1-k] = 2.5$$

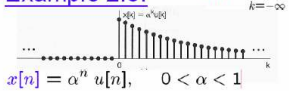
$$y[2] = \sum_{k=-\infty}^{+\infty} x[k]h[2-k] = 2.5$$

$$y[3] = \sum_{k=-\infty}^{+\infty} x[k]h[3-k] = 2.0$$

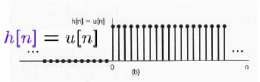
$y[n] = 0$ for $n > 3$

Example 2.3:

$y[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k]$ $\quad x[n] \longrightarrow \boxed{h[n]} \longrightarrow y[n]$

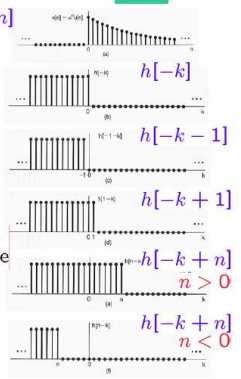


$x[n] = \alpha^n\, u[n], \quad 0 < \alpha < 1$



$h[n] = u[n]$

$x[n]$ $\quad h[-k]$ $\quad h[-k-1]$ $\quad h[-k+1]$ $\quad h[-k+n]$ $\quad n > 0$ $\quad h[-k+n]$ $\quad n < 0$



for $n < 0$, $\quad x[k]\, h[n-k] = 0 \;\Rightarrow\; y[n] = 0$

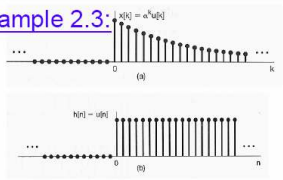
for $n \geq 0$, $\quad x[k]\, h[n-k] = \begin{cases} \alpha^k, & 0 \leq k \leq n \\ 0, & \text{otherwise} \end{cases}$

$$\Rightarrow y[n] = \sum_{k=0}^{n} \alpha^k = \frac{1-\alpha^{n+1}}{1-\alpha}$$

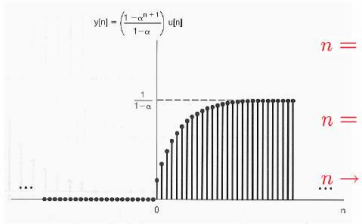
for all n, $\quad y[n] = \left(\frac{1-\alpha^{n+1}}{1-\alpha}\right) u[n]$

Example 2.3:

$x[n] \longrightarrow \boxed{h[n]} \longrightarrow y[n]$



$\alpha = \frac{7}{8}$

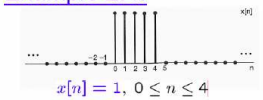


$n = 0 \quad y[0] = \dfrac{1-\frac{7}{8}}{1-\frac{7}{8}} = 1$

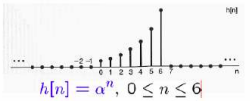
$n = 1 \quad y[1] = \dfrac{1-(\frac{7}{8})^2}{1-\frac{7}{8}} = \dfrac{15}{8}$

$n \to \infty \quad y[n] = \dfrac{1-0}{1-\frac{7}{8}} = 8$

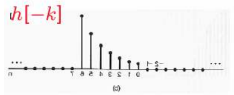
Example 2.4:



$x[n] = 1, \; 0 \leq n \leq 4$



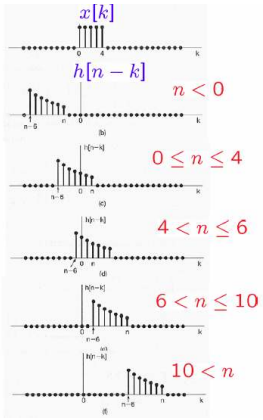
$h[n] = \alpha^n, \; 0 \leq n \leq 6$

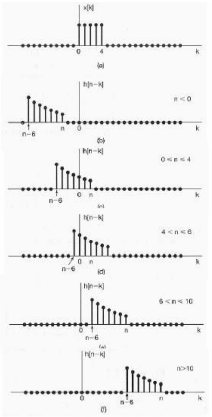


$h[-k]$

$x[n] \longrightarrow \boxed{h[n]} \longrightarrow y[n]$

$x[k]$ $\quad h[n-k]$ $\quad n < 0$ $\quad 0 \leq n \leq 4$ $\quad 4 < n \leq 6$ $\quad 6 < n \leq 10$ $\quad 10 < n$





$x[n] \longrightarrow \boxed{h[n]} \longrightarrow y[n]$

for $n < 0$, $\quad x[k]\, h[n-k] = 0 \;\Rightarrow\; y[n] = 0$

for $0 \leq n \leq 4$, $\quad x[k]\, h[n-k] = \begin{cases} \alpha^{n-k}, & 0 \leq k \leq n \\ 0, & \text{otherwise} \end{cases}$

$$\Rightarrow y[n] = \sum_{k=0}^{n} \alpha^{n-k} = \frac{1-\alpha^{n+1}}{1-\alpha}$$

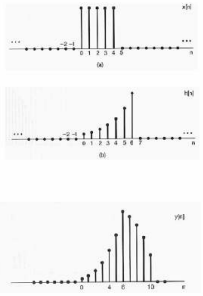
for $4 < n \leq 6$, $\quad x[k]\, h[n-k] = \begin{cases} \alpha^{n-k}, & 0 \leq k \leq 4 \\ 0, & \text{otherwise} \end{cases}$

$$\Rightarrow y[n] = \sum_{k=0}^{4} \alpha^{n-k} = \frac{\alpha^{n-4}-\alpha^{n+1}}{1-\alpha}$$

for $6 < n \leq 10$, $\quad x[k]\, h[n-k] = \begin{cases} \alpha^{n-k}, & (n-6) \leq k \leq 4 \\ 0, & \text{otherwise} \end{cases}$

$$\Rightarrow y[n] = \sum_{k=n-6}^{4} \alpha^{n-k} = \frac{\alpha^{n-4}-\alpha^{7}}{1-\alpha}$$

for $n > 10$, $\quad y[n] = 0$



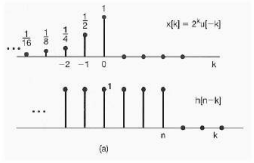
$x[n] \longrightarrow \boxed{h[n]} \longrightarrow y[n]$

$x[n] = 1, \ 0 \le n \le 4$

$h[n] = \alpha^n, \ 0 \le n \le 6$

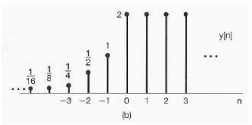
$$y[n] = \begin{cases} 0, & n < 0 \\ \frac{1-\alpha^{n+1}}{1-\alpha}, & 0 \le n \le 4 \\ \frac{\alpha^{n-4}-\alpha^{n+1}}{1-\alpha}, & 4 < n \le 6 \\ \frac{\alpha^{n-4}-\alpha^{7}}{1-\alpha}, & 6 < n \le 10 \\ 0, & 10 < n \end{cases}$$

- Example 2.5: $x[n] = 2^n u[-n]$, $h[n] = u[n]$ $\quad x[n] \longrightarrow \boxed{h[n]} \longrightarrow y[n]$



for $n \ge 0$, $y[n] = \sum_{k=-\infty}^{0} x[k]\,h[n-k] = \sum_{k=-\infty}^{0} 2^k$

$= \sum_{r=0}^{\infty} \left(\frac{1}{2}\right)^r = \frac{1}{1-(1/2)} = 2$



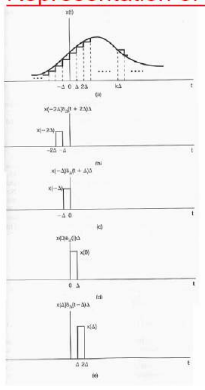
for $n < 0$, $y[n] = \sum_{k=-\infty}^{n} x[k]\,h[n-k] = \sum_{k=-\infty}^{n} 2^k$

$= \sum_{l=-n}^{\infty} \left(\frac{1}{2}\right)^l = \sum_{m=0}^{\infty} \left(\frac{1}{2}\right)^{m-n}$

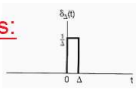
$= \left(\frac{1}{2}\right)^{-n} \sum_{m=0}^{\infty} \left(\frac{1}{2}\right)^m = 2^n \cdot 2 = 2^{n+1}$

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- Continuous-Time Linear Time-Invariant Systems
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- Properties of Linear Time-Invariant Systems
- Causal Linear Time-Invariant Systems Described by Differential & Difference Equations
- Singularity Functions

- Representation of CT Signals by Impulses:



$$\delta_\Delta(t) = \begin{cases} \frac{1}{\Delta}, & 0 \le t < \Delta \\ 0, & \text{otherwise} \end{cases}$$



$$\hat{x}(t) = \sum_{k=-\infty}^{+\infty} x(k\Delta)\delta_\Delta(t-k\Delta)\Delta$$

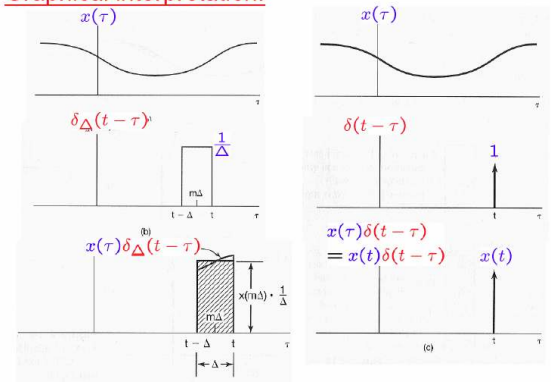
$$x(t) = \lim_{\Delta \to 0} \sum_{k=-\infty}^{+\infty} x(k\Delta)\delta_\Delta(t-k\Delta)\Delta$$

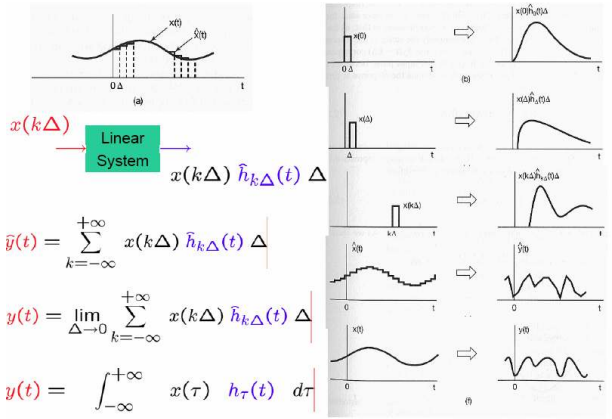
$$x(t) = \int_{-\infty}^{+\infty} x(\tau)\delta(t-\tau)d\tau$$

the sifting property of CT impulse

$x(t) =$ an integral of weighted, shifted impulses

- Graphical interpretation:





- CT Unit Impulse Response & Convolution Integral:

$$\delta(t-\tau) \longrightarrow \boxed{\text{Linear System}} \longrightarrow h_\tau(t)$$

$$x(t) \longrightarrow \boxed{\text{Linear System}} \longrightarrow y(t)$$

$$x(t) = \int_{-\infty}^{+\infty} x(\tau)\delta(t-\tau)d\tau$$

$$\Longrightarrow y(t) = \int_{-\infty}^{+\infty} x(\tau)h_\tau(t)d\tau$$

- If the linear system is also time-invariant $x(t) \longrightarrow \boxed{h(t)} \longrightarrow y(t)$
  - Then,
$$h_\tau(t) = h_0(t-\tau) = h(t-\tau)$$

- Hence, for an LTI system,

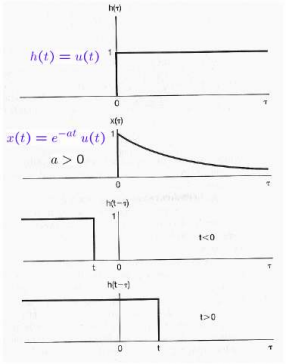
$$y(t) = \int_{-\infty}^{+\infty} x(\tau)h(t-\tau)d\tau \quad = \int_{-\infty}^{+\infty} h(\tau)x(t-\tau)d\tau$$

  - Known as the convolution of x(t) & h(t)
  - Referred as the convolution integral or the superposition integral

- Symbolically, $y(t) = x(t) * h(t) \quad = h(t) * x(t)$

- Example 2.6:

$$x(t) \longrightarrow \boxed{h(t)} \longrightarrow y(t)$$



for $t < 0$, $x(\tau)\, h(t-\tau) = 0$

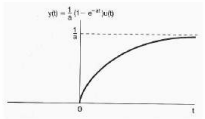
$\Rightarrow \quad y(t) = 0$

for $t \geq 0$, $x(\tau)\, h(t-\tau) = \begin{cases} e^{-a\tau}, & 0 \leq \tau \leq t \\ 0, & \text{otherwise} \end{cases}$

$$\Rightarrow \quad y(t) = \int_0^t e^{-a\tau} d\tau$$

$$= -\frac{1}{a} e^{-a\tau} \Big|_0^t$$

$$= \frac{1}{a}(1 - e^{-at})$$

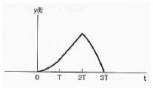


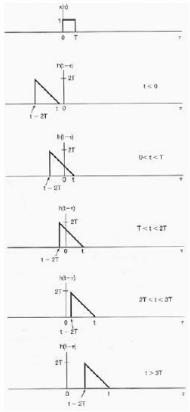
- Example 2.7: $x(t) \longrightarrow \boxed{h(t)} \longrightarrow y(t)$

$$x(t) = \begin{cases} 1, & 0 < t < T \\ 0, & \text{otherwise} \end{cases}$$

$$h(t) = \begin{cases} t, & 0 < t < 2T \\ 0, & \text{otherwise} \end{cases}$$

$$y(t) = \begin{cases} 0, & t < 0 \\ \frac{1}{2}t^2, & 0 < t < T \\ Tt - \frac{1}{2}T^2, & T < t < 2T \\ -\frac{1}{2}t^2 + Tt + \frac{3}{2}T^2, & 2T < t < 3T \\ 0, & 3T < t \end{cases}$$



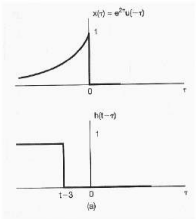


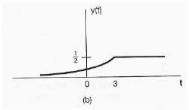
- Example 2.8:

$$x(t) = e^{2t} u(-t)$$

$$h(t) = u(t-3)$$

$$x(t) \longrightarrow \boxed{h(t)} \longrightarrow y(t)$$





for $t - 3 \leq 0$, $\quad y(t) = \int_{-\infty}^{t-3} e^{2\tau} d\tau$

$$= \frac{1}{2} e^{2(t-3)}$$

for $t - 3 \geq 0$, $\quad y(t) = \int_{-\infty}^{0} e^{2\tau} d\tau$

$$= \frac{1}{2}$$

- Discrete-Time Linear Time-Invariant Systems
  - The convolution sum
- Continuous-Time Linear Time-Invariant Systems
  - The convolution integral
- Properties of Linear Time-Invariant Systems
- Causal Linear Time-Invariant Systems Described by Differential & Difference Equations
- Singularity Functions

- Convolution Sum & Integral of LTI Systems:

$$y[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k] = x[k] * h[n]$$

$$y(t) = \int_{-\infty}^{+\infty} x(\tau)h(t-\tau)d\tau = x(t) * h(t)$$

$x[n] \rightarrow$ h[n] $\rightarrow y[n]$

$x(t) \rightarrow$ h(t) $\rightarrow y(t)$

- Properties of LTI Systems

1. Commutative property
2. Distributive property
3. Associative property
4. With or without memory
5. Invertibility
6. Causality
7. Stability
8. Unit step response

- Commutative Property: $n-k=r$

$$x[n] * h[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k] = \sum_{r=-\infty}^{+\infty} x[n-r]h[r]$$

$$= \sum_{r=-\infty}^{+\infty} h[r]x[n-r] = h[n] * x[n]$$

$$x(t) * h(t) = \int_{-\infty}^{+\infty} x(\tau)h(t-\tau)d\tau \qquad t-\tau=\sigma,\ -d\tau=d\sigma$$

$$= \int_{+\infty}^{-\infty} x(t-\sigma)h(\sigma)(-d\sigma) = \int_{-\infty}^{+\infty} x(t-\sigma)h(\sigma)d\sigma$$

$$= \int_{-\infty}^{+\infty} h(\sigma)x(t-\sigma)d\sigma = h(t) * x(t)$$

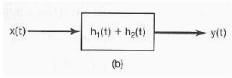
- Distributive Property:

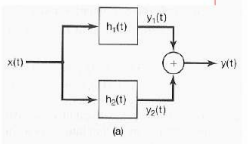
$$x[n] * h[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k]$$

$$x(t) * h(t) = \int_{-\infty}^{+\infty} x(\tau)h(t-\tau)d\tau$$

$$x[n] * \left(h_1[n] + h_2[n]\right) = x[n] * h_1[n] + x[n] * h_2[n]$$

$$x(t) * \left(h_1(t) + h_2(t)\right) = x(t) * h_1(t) + x(t) * h_2(t)$$





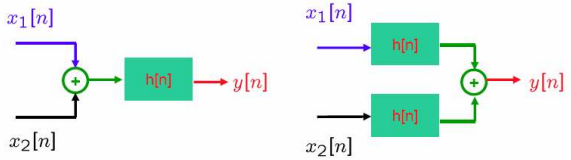
- **Distributive Property:**

$$x[n] * h[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k]$$

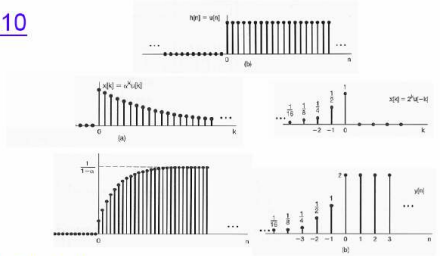
$$x(t) * h(t) = \int_{-\infty}^{+\infty} x(\tau)h(t-\tau)d\tau$$

$$\Big(x_1[n] + x_2[n]\Big) * h[n] = x_1[n] * h[n] + x_2[n] * h[n]$$

$$\Big(x_1(t) + x_2(t)\Big) * h(t) = x_1(t) * h(t) + x_2(t) * h(t)$$

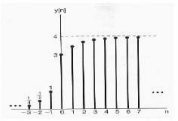


- **Example 2.10**



$$x[n] = \left(\frac{1}{2}\right)^n u[n] + 2^n u[-n]$$

$$y[n] = x[n] * h[n]$$
$$= \Big(x_1[n] + x_2[n]\Big) * h[n]$$
$$= x_1[n] * h[n] + x_2[n] * h[n]$$



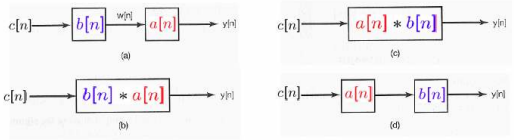
- **Associative Property:**

$$x[n] * h[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k]$$

$$x(t) * h(t) = \int_{-\infty}^{+\infty} x(\tau)h(t-\tau)d\tau$$

$$a[n] * \Big(b[n] * c[n]\Big) = \Big(a[n] * b[n]\Big) * c[n]$$

$$a(t) * \Big(b(t) * c(t)\Big) = \Big(a(t) * b(t)\Big) * c(t)$$



- **Memoryless:**
  - A DT LTI system is memoryless if $h[n] = 0$ for $n \neq 0$
  - The impulse response: $h[n] = K\delta[n], \quad K = h[0]$
  - The convolution sum: $y[n] = x[n] * h[n] = Kx[n]$
  - Similarly, for CT LTI system:

$$y(t) = x(t) * h(t) = Kx(t)$$

- Invertibility:

$x(t) \rightarrow$ h1(t) $\rightarrow y(t) \rightarrow$ h2(t) $\rightarrow w(t)$

$$y(t) = x(t) * h_1(t) \qquad w(t) = y(t) * h_2(t)$$

$$\Rightarrow w(t) = x(t) * h_1(t) * h_2(t)$$

$x(t) \rightarrow$ Identity System $\delta(t)$ $\rightarrow x(t)$

$$x(t) = x(t) * \delta(t)$$

$$\Longrightarrow h_2(t) * h_1(t) = \delta(t)$$

- Example 2.11: Pure time shift

$x(t) \rightarrow$ h1(t) $\rightarrow y(t) \rightarrow$ h2(t) $\rightarrow w(t)$

- $y(t) = x(t - t_0)$
  - delay if $t_0 > 0$
  - advance if $t_0 < 0$

$$\Rightarrow \quad h_1(t) = \delta(t - t_0) \quad \Rightarrow \quad x(t - t_0) = x(t) * \delta(t - t_0)$$

- $w(t) = x(t) = y(t + t_0)$

$$\Rightarrow \quad h_2(t) = \delta(t + t_0) \quad \Rightarrow \quad y(t + t_0) = y(t) * \delta(t + t_0)$$

$$\Rightarrow \quad h_1(t) * h_2(t) = \delta(t - t_0) * \delta(t + t_0) = \delta(t)$$

- Example 2.12

$x[n] \rightarrow$ h1[n] $\rightarrow y[n] \rightarrow$ h2[n] $\rightarrow w[n]$

$$h_1[n] = u[n]$$

$$\Rightarrow \; y[n] = \sum_{k=-\infty}^{\infty} x[k]\, u[n-k] \; = \sum_{k=-\infty}^{n} x[k]$$

$\Rightarrow$ a running-sum operation

- Its inverse is a first difference operation:

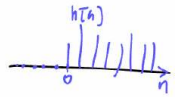
$$w[n] = y[n] - y[n-1] \;\Rightarrow\; h_2[n] = \delta[n] - \delta[n-1]$$

$$\Rightarrow \; h_1[n] * h_2[n] = u[n] - u[n-1] = \delta[n]$$

- Causality:
  - The output of a causal system depends only on the present and past values of the input to the system
  - Specifically, y[n] must not depend on x[k], for k > n

$h[n-k] = 0$, for $k > n$

$h[n] = 0$, for $n < 0$



  - It implies that the system is initially rest
  - A CT LTI system is causal if $h(t) = 0$, for $t < 0$

- Convolution Sum & Integral

$$y[n] = \sum_{k=-\infty}^{+\infty} x[k]\ h[n-k]$$

$$= \sum_{k=-\infty}^{n} x[k]\ h[n-k]$$

$$= \sum_{k=0}^{\infty} h[k]\ x[n-k]$$

$$y(t) = \int_{-\infty}^{\infty} x(\tau)\ h(t-\tau)\ d\tau$$

$$= \int_{-\infty}^{t} x(\tau)\ h(t-\tau)\ d\tau$$

$$= \int_{0}^{\infty} h(\tau)\ x(t-\tau)\ d\tau$$

- Stability:
  - A system is stable if every bounded input produces a bounded output

$$x[n] \rightarrow \boxed{\text{Stable LTI}} \rightarrow y[n]$$

$|x[n]| < B$ for all n $\quad |y[n]| = \left| \sum_{k=-\infty}^{+\infty} h[k]x[n-k] \right|$

$$\Rightarrow |y[n]| \le \sum_{k=-\infty}^{+\infty} |h[k]||x[n-k]|$$

$$\Rightarrow |y[n]| \le B \left(\sum_{k=-\infty}^{+\infty} |h[k]| \right)$$

if $\sum_{k=-\infty}^{+\infty} |h[k]| < \infty$ (absolutely summable) then, $y[n]$ is bounded

- Stability:
  - For CT LTI stable system:

$$x(t) \rightarrow \boxed{\text{Stable LTI}} \rightarrow y(t)$$

$|x(t)| < B$ for all t $\quad |y(t)| = \left| \int_{-\infty}^{+\infty} h(\tau)x(t-\tau)d\tau \right|$

$$\Rightarrow |y(t)| \le \int_{-\infty}^{+\infty} |h(\tau)||x(t-\tau)|d\tau$$

$$\Rightarrow |y(t)| \le B \left(\int_{-\infty}^{+\infty} |h[(\tau)|d\tau \right)$$

if $\int_{-\infty}^{+\infty} |h(\tau)|d\tau < \infty$ (absolutely integrable) then, $y(t)$ is bounded

- Example 2.13: Pure time shift
  - $y[n] = x[n-n_0]$ & $h[n] = \delta[n-n_0]$
  - $y(t) = x(t-t_0)$ & $h(t) = \delta(t-t_0)$

$\Rightarrow \sum_{n=-\infty}^{+\infty} |h[n]| = \sum_{n=-\infty}^{+\infty} |\delta[n-n_0]| = 1$ absolutely summable

$\Rightarrow \int_{-\infty}^{+\infty} |h(\tau)| = \int_{-\infty}^{+\infty} |\delta(\tau-t_0)|d\tau = 1$ absolutely integrable

$\Rightarrow$ A (CT or DT) pure time shift is stable

- Example 2.13: Accumulator
  - $y[n] = \sum_{k=-\infty}^{n} x[k]$ & $h[n] = u[n]$
  - $y(t) = \int_{-\infty}^{t} x(\tau)d\tau$ & $h(t) = u(t)$

$\Rightarrow \sum_{n=-\infty}^{+\infty} |h[n]| = \sum_{n=0}^{+\infty} |u[n]| = \infty$ NOT absolutely summable

$\Rightarrow \int_{-\infty}^{+\infty} |h(\tau)| = \int_{0}^{\infty} |u(\tau)|d\tau = \infty$ NOT absolutely integrable

$\Rightarrow$ A accumulator or integrator is NOT stable

- Unit Step Response:

$h[n] = \delta[n] * h[n]$

  - For an LTI system, its impulse response is:

$\delta[n] \rightarrow$ DT LTI $\rightarrow h[n]$ $\quad \delta(t) \rightarrow$ CT LTI $\rightarrow h(t)$

  - Its unit step response is:

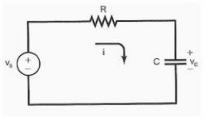
$u[n] \rightarrow$ DT LTI $\rightarrow s[n]$ $\quad u(t) \rightarrow$ CT LTI $\rightarrow s(t)$

$\Rightarrow s[n] = u[n] * h[n] = \sum_{k=-\infty}^{n} h[n]$ $\quad \Rightarrow s(t) = u(t) * h(t) = \int_{-\infty}^{t} h(\tau)d\tau$

$\Rightarrow h[n] = s[n] - s[n-1]$ $\quad \Rightarrow h(t) = \frac{ds(t)}{dt}$

- Discrete-Time Linear Time-Invariant Systems
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- Singularity Functions

- Linear Constant-Coefficient Differential Equations
  - e.x., RC circuit

Input signal: $v_s(t)$  Output signal: $v_c(t)$

$\Rightarrow \frac{dv_c(t)}{dt} + \frac{1}{RC}v_c(t) = \frac{1}{RC}v_s(t)$

$x(t) \rightarrow$ RC Circuit $\rightarrow y(t)$ $\quad \Rightarrow \frac{d}{dt}y(t) + a\,y(t) = b\,x(t)$

- **Linear Constant-Coefficient Differential Equations**
  - For a general CT LTI system,

$$x(t) \rightarrow \boxed{\text{CT LTI}} \rightarrow y(t)$$

$$a_N \frac{d^N}{dt^N} y(t) + a_{N-1} \frac{d^{N-1}}{dt^{N-1}} y(t) + \cdots + a_1 \frac{d}{dt} y(t) + a_0 y(t)$$

$$= b_M \frac{d^M}{dt^M} x(t) + b_{M-1} \frac{d^{M-1}}{dt^{M-1}} x(t) + \cdots + b_1 \frac{d}{dt} x(t) + b_0 x(t)$$

$$\Rightarrow \sum_{k=0}^{N} a_k \frac{d^k}{dt^k} y(t) = \sum_{k=0}^{M} b_k \frac{d^k}{dt^k} x(t)$$

$$\Rightarrow h(t) = ?$$

- **Linear Constant-Coefficient Difference Equations**
  - For a general DT LTI system,

$$x[n] \rightarrow \boxed{\text{DT LTI}} \rightarrow y[n]$$

$$a_0 y[n] + a_1 y[n-1] + \cdots + a_{N-1} y[n-N+1] + a_N y[n-N]$$

$$= b_0 x[n] + b_1 x[n-1] + \cdots + b_{M-1} x[n-M+1] + b_M x[n-M]$$

$$\Rightarrow \sum_{k=0}^{N} a_k \, y[n-k] = \sum_{k=0}^{M} b_k \, x[n-k]$$

$$\Rightarrow h[n] = ?$$

- **Recursive Equation:**

$$a_0 y[n] + a_1 y[n-1] + \cdots + a_{N-1} y[n-N+1] + a_N y[n-N]$$

$$= b_0 x[n] + b_1 x[n-1] + \cdots + b_{M-1} x[n-M+1] + b_M x[n-M]$$

$$\sum_{k=0}^{N} a_k \, y[n-k] = \sum_{k=0}^{M} b_k \, x[n-k]$$

$$\Rightarrow y[n] = \frac{1}{a_0} \left\{ \sum_{k=0}^{M} b_k \, x[n-k] - \sum_{k=1}^{N} a_k \, y[n-k] \right\}$$

- **Recursive Equation:**

$$\delta[n] \rightarrow \boxed{\text{LTI}} \rightarrow h[n]$$
$$x[n] \rightarrow \boxed{\text{LTI}} \rightarrow y[n]$$

  - For example, (Example 2.15)

$$y[n] - \frac{1}{2} y[n-1] = x[n] \qquad \begin{matrix} y[n] = 0, & \text{for } n \le -1 \\ x[n] = K\delta[n] \end{matrix}$$

$$\Rightarrow \begin{cases} y[0] = x[0] + \frac{1}{2} y[-1] = K \\ y[1] = x[1] + \frac{1}{2} y[0] = \frac{1}{2} K \\ y[2] = x[2] + \frac{1}{2} y[1] = \left(\frac{1}{2}\right)^2 K \\ \vdots \\ y[n] = x[n] + \frac{1}{2} y[n-1] = \left(\frac{1}{2}\right)^n K \end{cases}$$

$$\Rightarrow h[n] = \left(\frac{1}{2}\right)^n u[n]$$



$\Rightarrow$ an Infinite Impulse Response (IIR) system

- **Nonrecursive Equation:**
  - When N = 0,

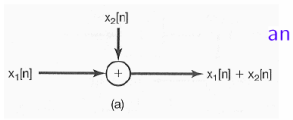
$$\Rightarrow\ y[n] = \sum_{k=0}^{M} \left(\frac{b_k}{a_0}\right) x[n-k]$$

$$\Rightarrow\ h[n] = \begin{cases} \frac{b_n}{a_0}, & 0 \le n \le M \\ 0, & \text{otherwise} \end{cases}$$

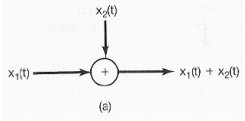
⇒ a Finite Impulse Response (FIR) system



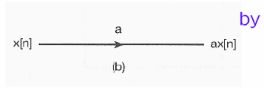
- **Block Diagram Representations:**

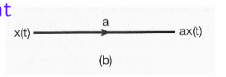


an adder

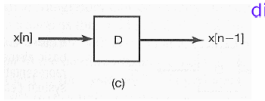


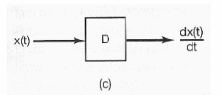
multiplication by a coefficient





a unit delay/ differentiator

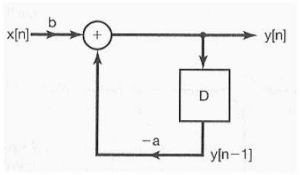




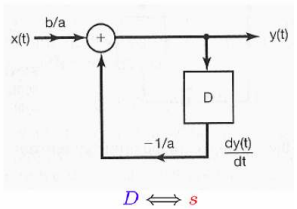
- **Block Diagram Representations:**

$$y[n] + ay[n-1] = bx[n] \qquad \frac{d}{dt}y(t) + ay(t) = bx(t)$$

$$y[n] = -ay[n-1] + bx[n] \qquad y(t) = -\frac{1}{a}\frac{d}{dt}y(t) + \frac{b}{a}x(t)$$



$D \Longleftrightarrow z^{-1}$



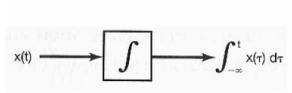
$D \Longleftrightarrow s$

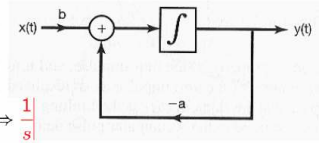
- **Block Diagram Representations:**

$$\frac{d}{dt}y(t) = bx(t) - ay(t)$$

$$\Rightarrow\ y(t) = \int_{-\infty}^{t} \left[bx(\tau) - ay(\tau)\right] d\tau$$

$$\Rightarrow\ y(t) = y(t_0) + \int_{t_0}^{t} \left[bx(\tau) - ay(\tau)\right] d\tau$$



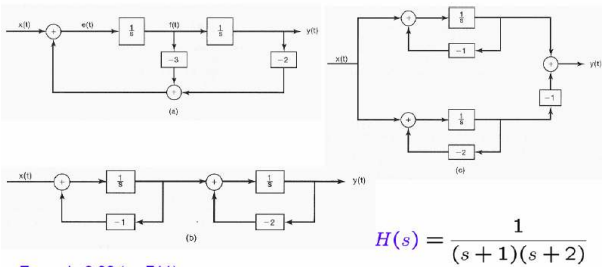


$\int \Longleftrightarrow \frac{1}{s}$

- Block Diagram Representations:

$$\int \Longleftrightarrow \frac{1}{s}$$

$$\frac{d^2}{dt^2}y(t) + 3\frac{d}{dt}y(t) + 2y(t) = x(t)$$

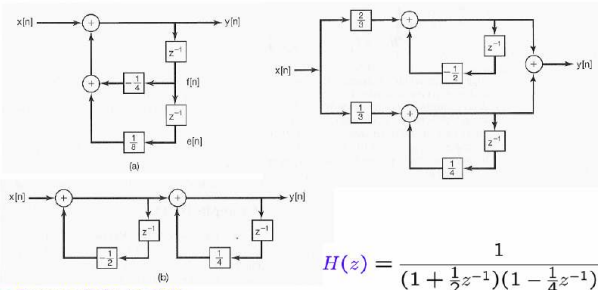


$$H(s) = \frac{1}{(s+1)(s+2)}$$

- Example 9.30 (pp.711)

- Block Diagram Representations:

$$y[n] + \frac{1}{4}y[n-1] - \frac{1}{8}y[n-2] = x[n]$$



$$H(z) = \frac{1}{(1+\frac{1}{2}z^{-1})(1-\frac{1}{4}z^{-1})}$$

- Example 10.30 (pp.786)

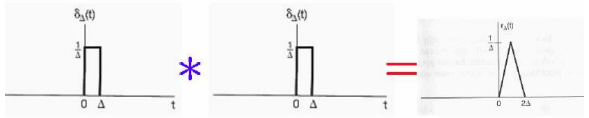
- Discrete-Time Linear Time-Invariant Systems
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- Singularity Functions

- Singularity Functions
  - CT unit impulse function is one of singularity functions

$$x(t) = x(t) * \delta(t)$$

$$\delta(t) = \delta(t) * \delta(t)$$

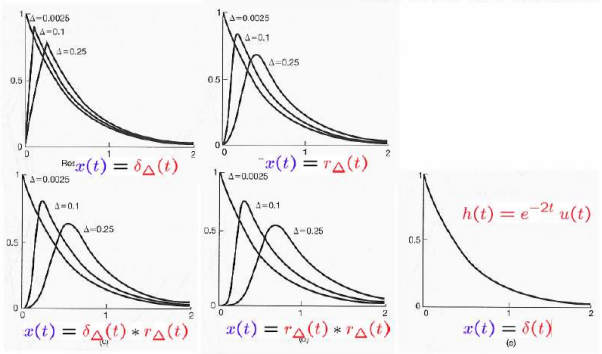
$$r_\Delta(t) = \delta_\Delta(t) * \delta_\Delta(t)$$



$$\lim_{\Delta \to 0} \delta_\Delta(t) = \delta(t) \quad \Rightarrow \lim_{\Delta \to 0} r_\Delta(t) = \delta(t)$$

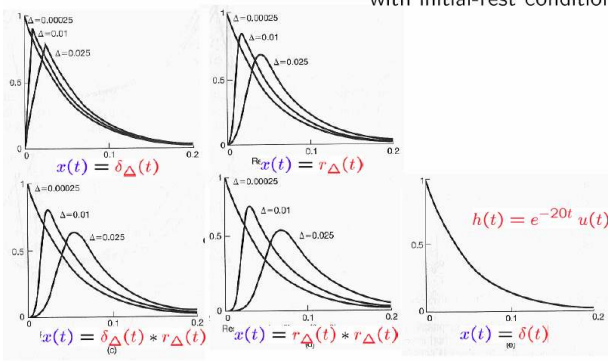
- Example 2.16 $\frac{d}{dt}y(t) + 2\,y(t) = x(t)$

with initial-rest condition



- Example 2.16 $\frac{d}{dt}y(t) + 20\,y(t) = x(t)$

with initial-rest condition



- Defining the Unit Impulse through Convolution:

$$x(t) = x(t) * \delta(t)$$

- Let $x(t) = 1$,

$$1 = x(t) = x(t) * \delta(t) = \delta(t) * x(t)$$
$$= \int_{-\infty}^{\infty} \delta(\tau)x(t-\tau)d\tau = \int_{-\infty}^{\infty} \delta(\tau)d\tau$$

- So that the unit impulse has unit area

- Defining Unit Impulse through Convolution:

- Alternatively, consdier an arbitrary signal $g(t)$,

$$g(-t) = g(-t) * \delta(t) = \int_{-\infty}^{\infty} g(\tau - t)\delta(\tau)d\tau$$

$$g(0) = \int_{-\infty}^{\infty} g(\tau)\delta(\tau)d\tau$$

- Define $x(t-\tau) = g(\tau)$

$$x(t) = g(0) = \int_{-\infty}^{\infty} g(\tau)\delta(\tau)d\tau$$
$$= \int_{-\infty}^{\infty} x(t-\tau)\delta(\tau)d\tau = x(t) * \delta(t)$$

Defining Unit Impulse through Convolution:

- Consider the signal $f(t)\delta(t)$

$$\int_{-\infty}^{\infty} g(\tau)f(\tau)\delta(\tau)d\tau = g(0)f(0)$$

- On the other hand, consider the signal $f(0)\delta(t)$

$$\int_{-\infty}^{\infty} g(\tau)f(0)\delta(\tau)d\tau = g(0)f(0)$$

- Therefore,

$$f(t)\delta(t) = f(0)\delta(t)$$

Unit Doublets of Derivative Operation:

- A system: Output is the derivative of input

$$y(t) = \frac{d}{dt}x(t)$$

$\Rightarrow$ The unit impulse response of the system is the derivative of the unit impulse, which is called the unit doublet $u_1(t)$

- That is, from $x(t) = x(t) * \delta(t)$, we have

$$\frac{d}{dt}x(t) = x(t) * u_1(t)$$

Unit Doublets of Derivative Operation:

- Similarly,

$$\frac{d^2}{dt^2}x(t) = x(t) * u_2(t)$$

- But,

$$\frac{d^2}{dt^2}x(t) = \frac{d}{dt}\left(\frac{d}{dt}x(t)\right) = \big(x(t) * u_1(t)\big) * u_1(t)$$

- Therefore,

$$u_2(t) = u_1(t) * u_1(t)$$

- In general,

$u_k(t)$, $k > 0$, the kth derivative of $\delta(t)$

$$u_k(t) = u_1(t) * \cdots * u_1(t)$$

Unit Doublets of Integration Operation:

- A system: Output is the integral of input

$$y(t) = \int_{-\infty}^{\infty} x(\tau)d\tau$$

- Therefore,

$$u(t) = \int_{-\infty}^{\infty} \delta(\tau)d\tau$$

- Hence, we have

$$x(t) * u(t) = \int_{-\infty}^{t} x(\tau)d\tau$$

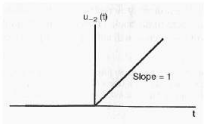
- Unit Doublets of Integration Operation:

  - Similarly,

$$u_{-2}(t) = u(t) * u(t) = \int_{-\infty}^{t} u(\tau) d\tau$$

  - That is,

$$u_{-2}(t) = t\, u(t)$$ the unit ramp function



- Unit Doublets of Integration Operation:

  - Moreover,

$$\begin{aligned} x(t) * u_{-2}(t) &= x(t) * u(t) * u(t) \\ &= \left(\int_{-\infty}^{t} x(\sigma) d\sigma \right) * u(t) \\ &= \int_{-\infty}^{t} \left(\int_{-\infty}^{\tau} x(\sigma) d\sigma \right) d\tau \end{aligned}$$

  - In general,

$$u_{-k}(t) = u(t) * \cdots * u(t) = \int_{-\infty}^{t} u_{-(k-1)}(\tau) d\tau$$

$$u_{-k}(t) = \frac{t^{k-1}}{(k-1)!} u(t)$$

- In Summary

$$\delta(t) = u_0(t)$$

$$u(t) = u_{-1}(t)$$

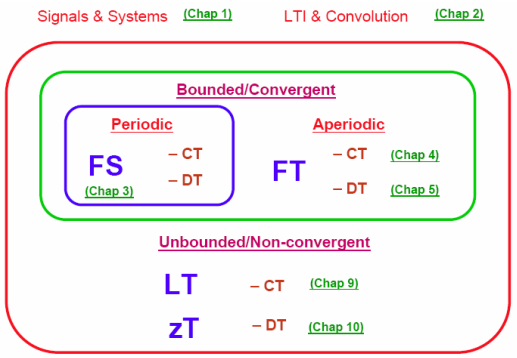
$$u_k(t)$$

$k > 0$,
Impulse response of a cascade of k differentiators

$k < 0$,
Impulse response of a cascade of $|k|$ integrators

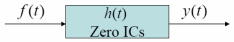
$$u(t) * u_1(t) = \delta(t) \quad \text{or,} \quad u_{-1}(t) * u_1(t) = u_0(t)$$

$$\Rightarrow \; u_k(t) * u_r(t) = u_{k+r}(t)$$



C-T Convolution Examples

Example 1:



Given : $f(t) = e^{t}u(-t) \quad h(t) = -\delta(t) + 2e^{-t}u(t)$

Find : $Zero-state\ response : y(t) = f(t) * h(t)$

Solution: $y(t) = f(t) * h(t) = f(t) * \left[-\delta(t) + 2e^{-t}u(t)\right]$

First we'll use properties of convolution to break this down into sub-problems.

By the distributive property we have:

call this $h_1(t)$

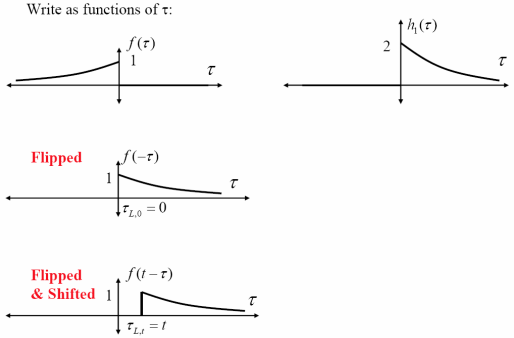
$$y(t) = -f(t) * \delta(t) + f(t) * 2e^{-t}u(t)$$

Use Property of Convolution with impulses

Call this convolution $y_1(t)$

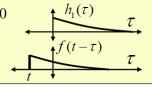
$$y(t) = -f(t) + y_1(t)$$

Write as functions of τ:



Only two cases for product $h_1(\tau)f(t-\tau)$:

RI: $t < 0$



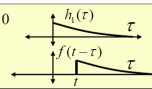
$$\Rightarrow y_1(t) = \int_0^{\infty}\left[e^{t-\tau}\right]2e^{-\tau}d\tau$$

$$= 2e^{t}\int_0^{\infty} e^{-2\tau}d\tau$$

$$= 2e^{t}\left[\frac{-1}{2}e^{-2\tau}\right]_0^{\infty} = -e^{t}\left[0-1\right]$$

$$\boxed{y_1(t) = e^{t} \quad for \quad t < 0}$$

RII: $t \geq 0$



$$y_1(t) = \int_t^{\infty}\left[e^{t-\tau}\right]2e^{-\tau}d\tau$$

$$= -e^{t}\left[e^{-2\tau}\right]_t^{\infty} = -e^{t}\left[0-e^{-2t}\right]$$

$$= e^{-t}$$

$$\boxed{y_1(t) = e^{-t} \quad for \quad t \geq 0}$$

Now… assemble these results together:

$$y(t) = -f(t) + y_1(t)$$

So:

$$y(t) = \begin{cases} \underbrace{-e^{t}}_{-f(t)} + \underbrace{e^{t}}_{y_1(t)} = 0, & for \;\; t < 0 \\ 0 + e^{-t} = e^{-t}, & for \;\; t \geq 0 \end{cases}$$

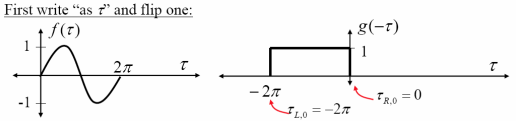
$$y(t) = \begin{cases} 0, & for \;\; t < 0 \\ e^{-t}, & for \;\; t \geq 0 \end{cases}$$

Example 2: Given: $f(t) = \sin(t)\left[u(t) - u(t-2\pi)\right]$

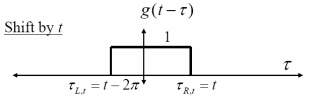
$g(t) = \left[u(t) - u(t-2\pi)\right]$

Find: $c(t) = f(t) * g(t)$

First write "as τ" and flip one:

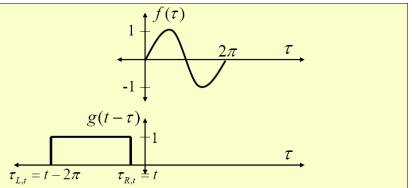


Shift by t



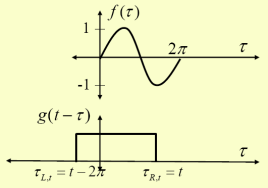
Now find the Regions of overlap… there are Four Regions

RI : $t \leq 0$

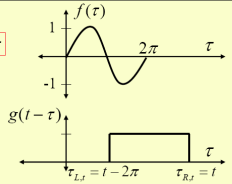


RII : $t > 0$, $t - 2\pi \leq 0$ $\Rightarrow 0 < t \leq 2\pi$

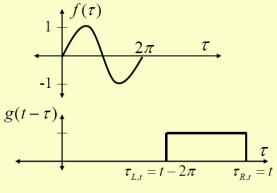
Note: this region includes the single t value at which you get complete overlap



RIII : $t - 2\pi \leq 2\pi$, $t > 2\pi$ $\Rightarrow 2\pi < t \leq 4\pi$



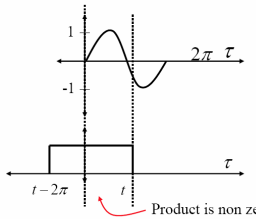
RIV : $t - 2\pi > 2\pi$ $\Rightarrow t > 4\pi$



RI and RIV Integrals

No Overlap ⇒Integrand = 0 ⇒ $c(t) = 0$ for $t \le 0$ and $t > 4\pi$

RII Integral

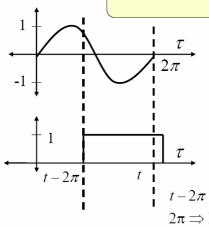


Product is non zero only here
⇒integrate 0 to t

$$\int_0^t 1\sin(\tau)d\tau = \left[-\cos(\tau)\right]_0^t = -\left[\cos(t)-1\right]$$
$$= 1-\cos(t) \quad for \quad 0 < t \le 2\pi$$

RIII Integral

Comment: note that you do not get a sinusoid out, because you only put part of a sinusoid in!

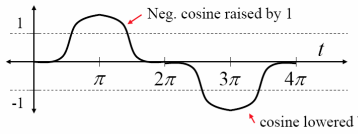


$t - 2\pi$

$2\pi \Rightarrow$ Integrate (t - 2π) to 2π

$$\int_{(t-2\pi)}^{2\pi} 1\sin(\tau)d\tau = \left[-\cos(\tau)\right]_{t-2\pi}^{2\pi} = -\left[\cos(2\pi)-\cos(t-2\pi)\right]$$
$$= \underbrace{\cos(t)-1}_{\text{cosine lowered by 1}} \quad for \quad 2\pi < t \le 4\pi \qquad \underbrace{= \cos(t)}_{\text{by periodicity}}$$

Now… assemble the parts that make up the output:

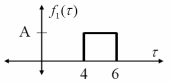


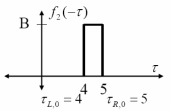
$$c(t) = \begin{cases} 0, & t \le 0 \\ 1-\cos t, & 0 < t \le 2\pi \\ \cos t - 1, & 2\pi < t \le 4\pi \\ 0, & t > 4\pi \end{cases}$$

Example 3:

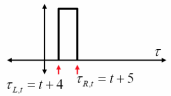
Given: $f_1(t) = A\left[u(t-4)-u(t-6)\right]$
$f_2(t) = \left[u(t+5)-u(t+4)\right]$
Find: $c(t) = f_1(t) * f_2(t)$

Write "as τ" and flip one:

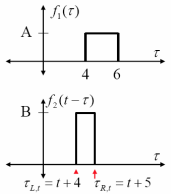




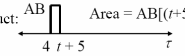
"Shift by t"



RI: $t + 5 < 4 \Rightarrow t < -1$ prod. = 0 ⇒ $c(t) = 0 \quad \forall\, t < -1$

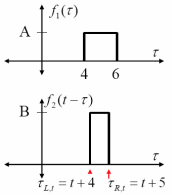


RII: $\left.\begin{matrix} t+5 \geq 4 \\ t+4 < 4 \end{matrix}\right\}$ $-1 \leq t < 0$

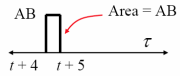
product: 

Area = AB[(t+5)-4] = ABt + AB

$$c(t) = \int_4^{t+5} AB\,d\tau = \boxed{ABt + AB \quad for \quad -1 \leq t < 0}$$

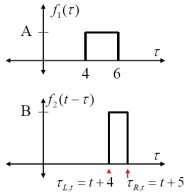


RIII: $\left.\begin{matrix} t+4 \geq 4 \\ t+5 < 6 \end{matrix}\right\}$ $0 \leq t < 1$

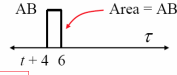
product: 

$$c(t) = \int_{t+4}^{t+5} AB\,d\tau$$

$$\boxed{c(t) = AB \quad for \quad 0 \leq t < 1}$$

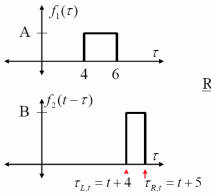


RIV: $\left.\begin{matrix} t+5 \geq 6 \\ t+4 < 6 \end{matrix}\right\}$ $1 \leq t < 2$

product: 

= 2AB - ABt

$$\boxed{c(t) = \int_{t+4}^{6} AB\,d\tau = 2AB - ABt \quad for \quad 1 \leq t < 2}$$

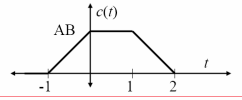


RV: $t+4 \geq 6 \Rightarrow t \geq 2$ product = 0

$$\boxed{c(t) = 0 \quad for \quad t \geq 2}$$

Now… assemble the pieces:

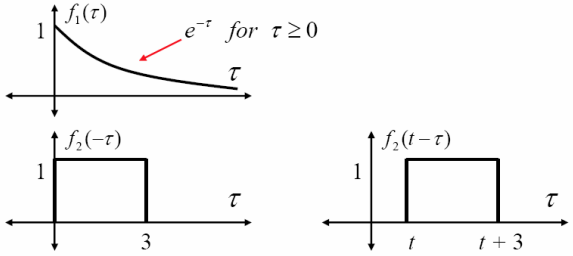
$$c(t) = \begin{cases} 0, & t < -1 \\ ABt + AB, & -1 \leq t < 0 \\ AB, & 0 \leq t < 1 \\ 2AB - ABt, & 1 \leq t < 2 \\ 0, & t \geq 2 \end{cases}$$



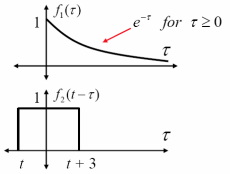
Example 4: Given : $f_1(t) = e^{-t} * u(t)$

$f_2(t) = u(t+3) - u(t)$

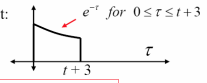
Find : $c(t) = f_1(t) * f_2(t)$



RI: $t + 3 < 0 \Rightarrow t <$ -3 prod = 0 $\Rightarrow$ $c(t) = 0$ $t <$ -3

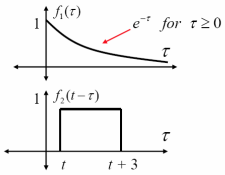


RII: $t+3 \geq 0$, $t < 0$ $\quad -3 \leq t < 0$ product:

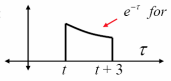


$\Rightarrow c(t) = \int_0^{t+3} e^{-\tau} d\tau = -\left[e^{-\tau}\right]_0^{t+3}$

$= -\left[e^{-(t+3)} - 1\right] =$ $1 - e^{-(t+3)}$ for $-3 \leq t < 0$



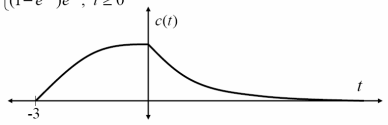
RIII: $t \geq 0$ product:



$\Rightarrow c(t) = \int_t^{t+3} e^{-\tau} d\tau = -\left[e^{-\tau}\right]_t^{t+3} = e^{-t} - e^{-(t+3)}$

$= e^{-t} - e^{-3}e^{-t} =$ $\left(1 - e^{-3}\right)e^{-t}$ for $t \geq 0$

$$c(t) = \begin{cases} 0, & t < -3 \\ 1 - e^{-(t+3)}, & -3 \leq t < 0 \\ (1 - e^{-3})e^{-t}, & t \geq 0 \end{cases}$$



Graphical Convolution Steps

Can do convolution this way when signals are know numerically or by equation

- Convolution involves the sum of a product of two signals: $x[i]h[n-i]$
- At each output index n, the product changes

"Commutativity" says we can flip either $x[i]$ or $h[i]$ and get the same answer

Step 1: Write both as functions of i: $x[i]$ & $h[i]$

Step 2: Flip $h[i]$ to get $h[-i]$ (The book calls this "fold")

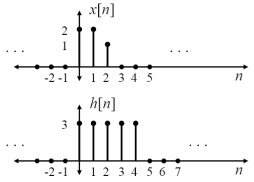
Repeat for each *n*

Step 3: For each output index n value of interest, shift by n to get $h[n - i]$

(Note: positive n gives right shift!!!!)

Step 4: Form product $x[i]h[n-i]$ and sum its elements to get the number $y[n]$

Example of Graphical Convolution



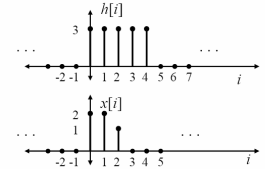
Find $y[n]=x[n]*h[n]$ for all integer values of n

Solution

For this problem I choose to flip $x[n]$

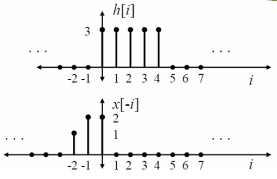
My personal preference is to flip the shorter signal although I sometimes don't follow that "rule"… only through lots of practice can you learn how to best choose which one to flip.

Step 1: Write both as functions of i: $x[i]$ & $h[i]$



Step 2: Flip $x[i]$ to get $x[-i]$

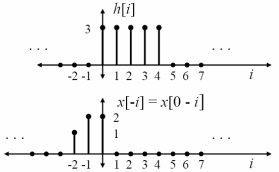
"Commutativity" says we can flip either $x[i]$ or $h[i]$ and get the same answer… **Here I flipped $x[i]$**



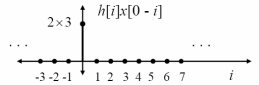
We want a solution for $n = \ldots -2, -1, 0, 1, 2, \ldots$ so must do Steps 3&4 for all n.

But… let's first do: **Steps 3&4 for $n = 0$** and then proceed from there.

Step 3: For $n = 0$, shift by n to get $x[n - i]$ ← For $n = 0$ case there is no shift! $x[0 - i] = x[-i]$



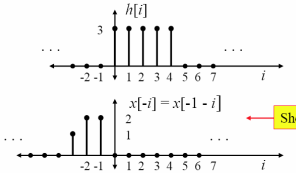
Step 4: For $n = 0$, Form the product $x[i]h[n - i]$ and sum its elements to give $y[n]$



Sum over i ⇒ $y[0] = 6$

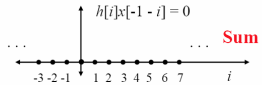
Steps 3&4 for all $n < 0$

Step 3: For $n < 0$, shift by n to get $x[n - i]$ ← Negative n gives a left-shift



Shown here for $n = -1$

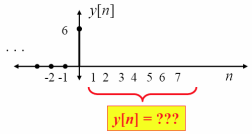
Step 4: For $n < 0$, Form the product $x[i]h[n - i]$ and sum its elements to give $y[n]$



Sum over i ⇒ $y[n] = 0 \quad \forall n < 0$

So… what we know so far is that:

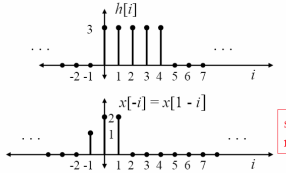
$$y[n] = \begin{cases} 0, & \forall n < 0 \\ 6, & n = 0 \end{cases}$$



So now we have to do Steps 3&4 for $n > 0$…

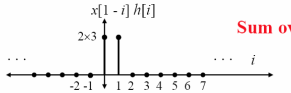
Steps 3&4 for $n = 1$

Step 3: For $n = 1$, shift by n to get $x[n - i]$ ← Positive n gives a Right-shift



shifted to the right by one

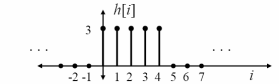
Step 4: For $n = 1$, Form the product $x[i]h[n - i]$ and sum its elements to give $y[n]$

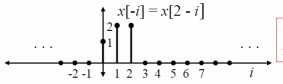


Sum over i ⇒ $y[1] = 6 + 6 = 12$

Steps 3&4 for $n = 2$

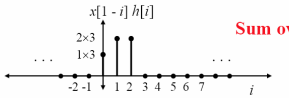
Step 3: For $n = 2$, shift by n to get $x[n - i]$ ← Positive n gives a Right-shift





shifted to the right by two

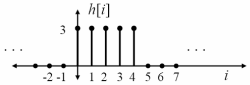
Step 4: For $n = 2$, Form the product $x[i]h[n - i]$ and sum its elements to give $y[n]$



Sum over i ⇒ $y[2] = 3 + 6 + 6 = 15$

Steps 3&4 for $n = 3$

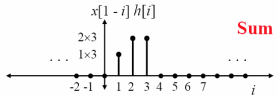
Step 3: For $n = 3$, shift by n to get $x[n - i]$ ← Positive n gives a Right-shift





shifted to the right by three

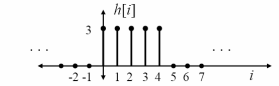
Step 4: For $n = 3$, Form the product $x[i]h[n - i]$ and sum its elements to give $y[n]$

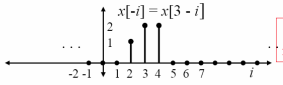


Sum over i ⇒ $y[3] = 3 + 6 + 6 = 15$

Steps 3&4 for $n = 4$

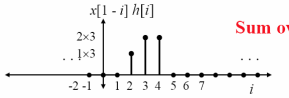
Step 3: For $n = 4$, shift by n to get $x[n - i]$ ← Positive n gives a Right-shift





shifted to the right by four

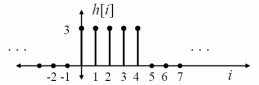
Step 4: For $n = 4$, Form the product $x[i]h[n - i]$ and sum its elements to give $y[n]$

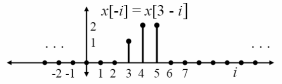


Sum over i ⇒ $y[4] = 3 + 6 + 6 = 15$

Steps 3&4 for $n = 5$

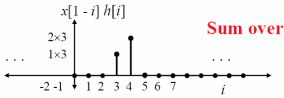
Step 3: For $n = 5$, shift by n to get $x[n - i]$ ← Positive n gives a Right-shift





shifted to the right by five

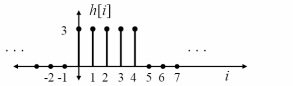
Step 4: For $n = 5$, Form the product $x[i]h[n - i]$ and sum its elements to give $y[n]$

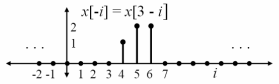


Sum over i ⇒ $y[5] = 3 + 6 = 9$

Steps 3&4 for $n = 6$

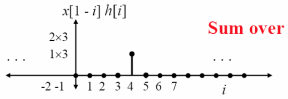
Step 3: For $n = 6$, shift by n to get $x[n - i]$ ← Positive n gives a Right-shift





shifted to the right by six

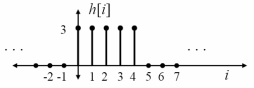
Step 4: For $n = 6$, Form the product $x[i]h[n - i]$ and sum its elements to give $y[n]$

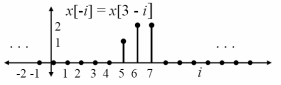


Sum over i ⇒ $y[6] = 3$

Steps 3&4 for all $n > 6$

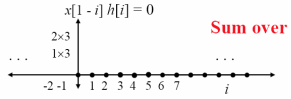
Step 3: For $n > 6$, shift by n to get $x[n - i]$ ← Positive n gives a Right-shift





shifted to the right by seven

Step 4: For $n > 6$, Form the product $x[i]h[n - i]$ and sum its elements to give $y[n]$

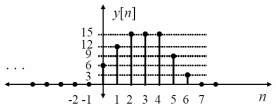


Sum over i ⇒ $y[n] = 0 \quad \forall n > 6$

So… now we know the values of $y[n]$ for all values of n

We just need to put it all together as a function…

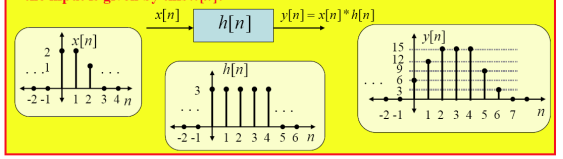
Here it is easiest to just plot it… you could also list it as a table.



Note that convolving these kinds of signals gives a "ramp-up" at the beginning and a "ramp-down" at the end.

Various kinds of "transients" at the beginning and end of a convolution are common.

BIG PICTURE: So… what we have just done is found the zero-state output of a system having an impulse response given by this $h[n]$ when the input is given by this $x[n]$:



Link: Web Demos of Graphical D-T Convolution

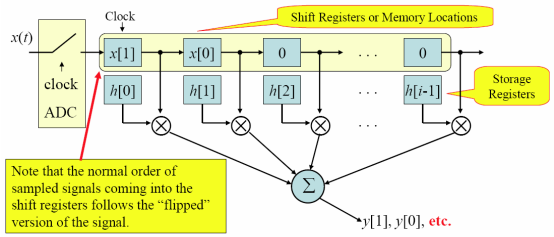
This is a good site that provides insight into how to visualize D-T convolution…

However, be sure you can do graphical convolution by hand without the aid of this site!!

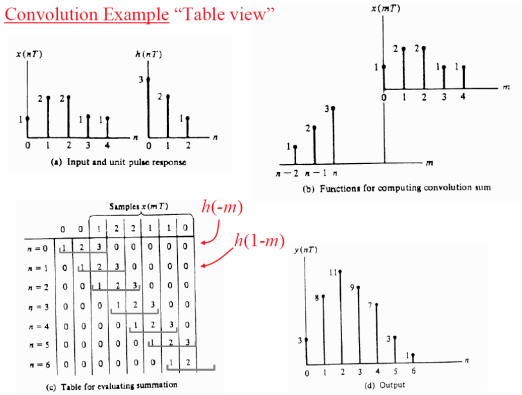
Implementation Issues

Consider a D-T system with impulse response $h[n]$ that has finite duration…

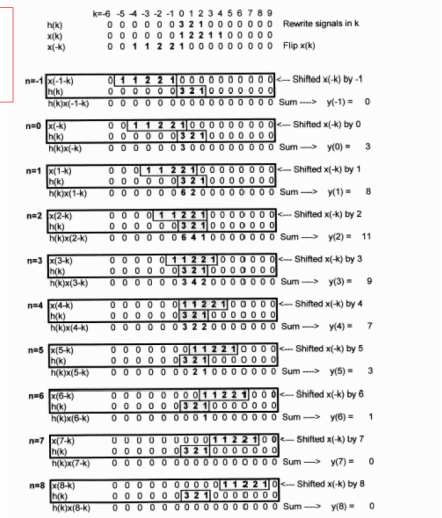
Could Build a digital ***hardware*** system or a ***software*** program for D-T convolution like this:



Convolution Example "Table view"

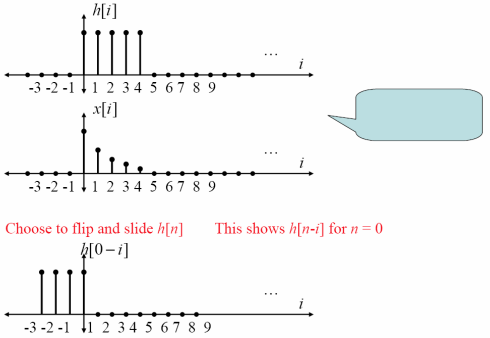


Discrete-Time Convolution Example: "Sliding Tape View"



D-T Convolution Examples

$x[n] = \left(\frac{1}{2}\right)^n u[n]$ $\quad$ $h[n] = u[n] - u[n-4]$

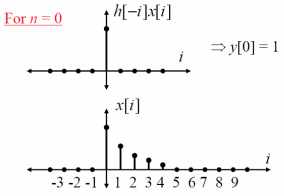


Choose to flip and slide $h[n]$ $\quad$ This shows $h[n\text{-}i]$ for $n = 0$

For $n < 0$ $\quad h[n-i]x(i) = 0 \quad \forall i$

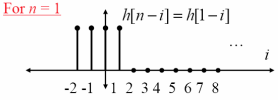
$\Rightarrow \boxed{y[n] = 0 \quad for \quad n < 0}$

For $n = 0$



$\Rightarrow y[0] = 1$

For $n = 1$



$\Rightarrow y(1) = 1 + ½ = 3/2$

Notice that for $n = 0$, $n = 1$, …, $n = 3$

The general result is:

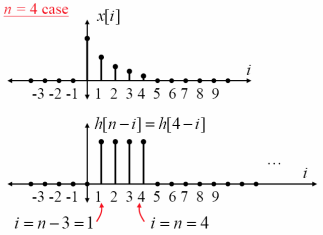
$$y[n] = \sum_{i=0}^{n} \left(\tfrac{1}{2}\right)^i \qquad for \quad n = 0,1,2,3$$

$$= \frac{1 - \left(\tfrac{1}{2}\right)^{n+1}}{1 - \tfrac{1}{2}} \qquad (Geometric \ \ Sum)$$

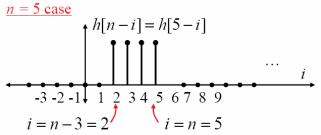
$$\boxed{y[n] = 2\left[1 - \left(\tfrac{1}{2}\right)^{n+1}\right] \qquad for \quad n = 0,1,2,3}$$

Now for $n = 4$, $n = 5$, …

$n = 4$ case



$n = 5$ case



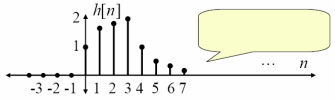
Notice that: for $n = 4, 5, 6, \ldots$

$$y[n] = \sum_{i=n-3}^{n} \left(\tfrac{1}{2}\right)^i \qquad for \quad n = 4,5,6,...$$

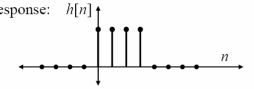
$$= \frac{\left(\tfrac{1}{2}\right)^{n-3} - \left(\tfrac{1}{2}\right)^{n+1}}{1 - \tfrac{1}{2}} \qquad then \ simplify!$$

Then we can write out the solution as:

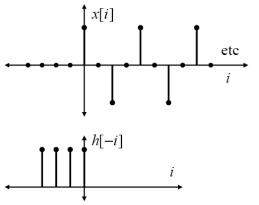
$$y[n] = \begin{cases} 0, & n < 0 \\ 2\left[1 - (1/2)^{n+1}\right], & n = 0,1,2,3 \\ 2\left[(1/2)^{n-3} - (1/2)^{n+1}\right], & n = 4,5,6,... \end{cases}$$



2. Same Impulse Response: $h[n]$



$x[n] = \cos\left(\tfrac{\pi}{2} n\right) h[n]$



Again $y[n] = 0$ for $n < 0$:

$y[0] = 1$

$y[1] = 1 + 0 = 1$

$y[2] = 1 + 0 - 1 = 0$

$y[3] = 1 + 0 - 1 + 0 = 0$

$y[4] = 0 - 1 + 0 + 1 = 0$

$y[5] = -1 + 0 + 1 + 0 = 0$

Notice: $y[n] = 0 \quad \forall \ n = 2, 3, 4, 5, \ldots!$

